RELATIVELY PRIME POLYNOMIALS ARE EXACTLY THOSE WITH NO COMMON ROOT IN AN ALGEBRAIC CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This lemma is the coprimality criterion quoted by lemma-recognize-separable (delivered card L48365) throughout the separability theory.

Editorial note. The label lemma-relatively-prime-polynomials names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter section preamble immediately preceding the lemma (source lines 1292-1315): over an algebraically closed field K, two polynomials are relatively prime exactly when they have no common root, because the maximal ideals of K[x] are the (x - alpha)
- Linear algebra: a system of linear equations with coefficients in k has a solution over k exactly when it has a solution over any extension field of k (solvability is a rank condition)

Lemma 0.1. *Two polynomials in $k[x]$ are relatively prime precisely when they have no common roots in an algebraic closure $\overline{k}$ of k.*

Proof. The claim is that any two polynomials P, Q generate (1) in $k[x]$ if and only if they generate (1) in $\overline{k}[x]$. This is a piece of linear algebra: a system of linear equations with coefficients in k has a solution if and only if it has a solution in any extension of k. Consequently, we can reduce to the case of an algebraically closed field, in which case the result is clear from what we have already proved. □

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